

CNN-LSTM based Approach for DDoS Detection

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Abstract—Distributed Denial of Service (DDoS) attacks have become increasingly common, causing financial and reputational losses for organizations. Despite the existence of numerous conventional detection solutions, DDoS attacks continue to rise in frequency, demanding effective models to detect and prevent them. This paper focuses on developing a machine learning-based approach for DDoS attack detection. By leveraging the power of machine learning, we aim to overcome the limitations of existing methods and propose a novel solution. Our work emphasizes the importance of exploring advanced models and techniques to enhance detection accuracy and efficiency. Through rigorous experimentation, we demonstrate the effectiveness of our approach in proactive defense against real-world DDoS attacks.

Keywords—DDOS, Detection, Machine learning

I. INTRODUCTION

Denial of Service attack (DoS) is an attack launched against networks and system resources to yield them unavailable to legitimate users. Therefore, this attack targets the availability of the systems, which is one of the essential things to protect. DoS attacks target web servers, default gateways, personal computers, etc., mostly by exploiting TCP/IP stack vulnerabilities. Some classic examples of DoS attacks are TCP Syn Flood, UDP Flood, ICMP Flood, Smurf, incomplete HTTP requests, etc. When multiple hosts coordinate to flood the victim with an abundance of attack packets simultaneously, it is called a Distributed Dos (DDoS) attack [1].

DDoS attacks may compromise online connected devices to generate a huge amount of network traffic targeting known services. These attacks are considered effective due to the ability to use multiple devices (botnets) infiltrated as a source of traffic to attack systems [1].

Machine Learning (ML) algorithms are considered an effective approach to detecting and preventing DDoS

attacks. They work by building models to differentiate normal and abnormal network traffic. The built model is then used to detect anomalies in the network. ML techniques can be model-driven, based on unsupervised or supervised algorithms. Supervised algorithms use historical data or labeled data to build the model. In contrast, unsupervised algorithms build the model by correlating upcoming data or putting new data into similar clusters. ML algorithms can be applied to DDoS detection and have been found to perform better than "signature-based" detection [2]. Beside ML algorithms, other DDoS prevention techniques are proposed [3]. An ingress filtering mechanism is used to monitor and restrict network traffic by allowing only legitimate traffic and blocking unauthorized traffic. Another mechanism used is Egress filtering. This mechanism monitors and restricts traffic that attempts to leave the network. This usually is done by a firewall that blocks packets that fail to meet certain security requirements. Moreover, History-based IP filtering uses a pre-built IP address dataset, and the edge router recognizes incoming packets accordingly.

Finally, Honeypots lure attackers by allowing them to attack the honeypots and not the actual system. They also help in obtaining the attacker's information by storing their logs, the type of attack and the type of program used. As for DDoS mitigation, many techniques were proposed [3]. Differentiated services is a technology based on the type of service byte in the IP header and is used to classify and manage network traffic and provide Quality of Service (QoS) on modern IP networks. Administrators use the Differentiated Services Code Point field in the IP packet address to distinguish packets, which contain 64 features at the traffic class level. They are used to mitigate DDOS attacks by limiting traffic so that matching criteria can identify each type of volumetric attack currently observed in the wild. Secondly, Class-Based Queuing (CBQ) is another technique used that is based on the concept of Class Traffic (COT), which is a mutually exclusive subset of

traffic. A certain amount of outbound bandwidth can then be allocated to traffic classes, so each connection point can be restricted on the bandwidth to share the link.

CBQ showed maintenance of QoS during a DDoS attack on groups of web servers. Resource Pricing also proposes a distributed gateway architecture and a payment protocol that charges dynamically variable prices on network, server, and information resources. It identifies clients by allotting a priority mechanism to desirable clients and punishing clients who cause a load on the server. Finally, Throttling is a technique used in which traffic that passes through the router to the source is limited by the throttle rate. Only the strong flows that do not respect their quota rates are penalized and no other flows. One of the most recent promising techniques to mitigate DDoS attacks is Moving Target Defense (MTD). MTD is an emerging defensive technology used to enhance network security by making it difficult for attackers to know their targeted node (i.e., Web Application Server). Hence, it promotes instability and randomness by modifying the system's attack surface to confuse the attackers, increasing the cost and effort required to conduct such attacks [4].

This research mainly focuses on developing a machine-learning technique to detect DDoS attacks.

II. RELATED WORK

Supervised and unsupervised learning for DDoS detection and the performance of Botnet DDOS attack detection using machine learning techniques was presented in [5]. Support Vector Machine (SVM), Artificial Neural Network (ANN), Naïve Bayes (NB), Decision Tree (DT), and Unsupervised Learning (USML) were used in the experiments. The experiments show that the accuracy for Unsupervised Machine learning was the highest, followed by the Decision tree and Support Vector Machine. The main issue is the unknown size of the used sample.

A botnet detection framework with sequential architecture based on machine learning was proposed for detecting attacks on IoT environments [6]. The correlated-feature selection method was adopted to decrease the unrelated features. The proposed model reached an accuracy of 99% in detecting DDOS attacks. The results show that ANN reaches the same accuracy as the hybrid model in multiple tests [7].

Likewise, Machine learning techniques were used to detect malware due to their high accuracy in detecting malware behavior. K-nearest neighbors, random forests, support vector machines and deep neural networks were used to differentiate between normal flows and DDoS attacks. The honeypot framework was used to detect malware installation attempts into the IoT device. Honeypot trains the model to learn the different malware traffic flows and activities. A key drawback was that the results of machine learning techniques were not

presented, in addition to the lack of information about the used dataset [2].

In [1], a hybrid intrusion detection system for DDoS attacks based on clustering and classification techniques is presented. To identify DDoS attacks, the authors grouped related network traffic flows using the K-means clustering technique and then classified these flows as either normal or attack traffic using SVMs. The suggested hybrid IDS detected DDoS assaults with a 96.43% accuracy rate.

A machine learning approach for DDoS detection in software-defined networks is presented in [3]. The authors classified network traffic flows as either normal or attack traffic using a mix of SVM, Random Forest, and k-Nearest Neighbor (KNN) algorithms. The suggested method detected DDoS assaults with a 99 % accuracy rate.

DDoS attack detection with feature selection using machine learning algorithms is presented in [4]. This study presented a DDoS attack detection system that uses feature selection methods to find the most pertinent features for categorization. The authors classified network traffic flows as either normal or attack traffic using SVM, Naive Bayes, and Decision Tree methods. The suggested method detected DDoS assaults with a 98.9% accuracy rate. Moreover, a machine learning-based intelligent DDoS assault detection system is presented in [8]. In this research, they presented an intelligent method for detecting DDoS attacks. In order to categorize network traffic flows as either regular or attack traffic, the authors employed the SVM algorithm. The suggested system's DDoS attack detection accuracy was 99 %.

III. BACKGROUND

A. DDoS attacks

DOS is an abbreviation for Denial of Service, which means that blocking a specific service leads to suspending that service. For example, stopping the HTTP service on a web server means stopping browsing the sites, and stopping the FTP service leads to the suspension of the FTP connection service on the server.

A DOS attack is executed using several techniques, including massive traffic or exploiting vulnerabilities such as the Apache http Remote Denial of Service (memory exhaustion) vulnerability. DDoS is an abbreviation for Distributed Denial of Service. These attacks can be divided into three main categories: Volume based attacks, which include ICMP floods, UDP floods, and other spoofed packet attacks. Protocol-based attacks include SYN floods, fragmented packet attacks, Ping of death, Smurf attacks and more. Application layer-based attacks, which include attacks like Zero-day attacks [1].

B. Machine Learning for DDoS Detection

Many algorithms are used to analyze and detect a DDoS attack, such as the Machine Learning algorithm, which helps detect the attack. Machine learning software aims to train a computer to perform specific tasks on its own.

Deep learning is a form of machine learning, which is a branch of artificial intelligence. Machine learning relies on algorithms that can be fed with structured data and analyzed to reach conclusions, while deep learning is characterized by the presence of different levels of algorithms, which form Artificial Neural Networks (ANN), which can understand unstructured data [5].

Supervised and unsupervised learning are machine-learning techniques used to solve a class of tasks through learning from experience and measuring performance.

Supervised learning is the learning in which we teach or train a machine using labeled data. Then, the machine is provided with a new set of examples to analyze and produce a valid classification. Conversely, unsupervised learning is intended to train the machine using unlabeled information by allowing the algorithm to learn from that information without direction. The task of the machine here is to learn and classify according to similarities, patterns and differences that exist in the data [5].

IV. METHODOLOGY

The methodology is designed to effectively detect and classify network traffic, particularly focusing on distinguishing between normal and malicious traffic, specifically Distributed Denial of Service (DDoS) attacks. The methodology comprises three main stages: finding a suitable dataset, traffic analysis using a machine learning (ML) module and updating the ML module. Each stage plays a crucial role in achieving accurate and up-to-date detection results.



Fig. 2. Research Methodology

Finding a suitable dataset to ensure the representation of various forms of network traffic, including both normal and attack traffic, the initial step of this stage involves thorough dataset acquisition and selection. We meticulously search and gather datasets encompassing diverse network scenarios, traffic patterns, and attack types. We aim to enhance the model's ability to generalize and effectively identify anomalous behavior by including a wide range of data.

Traffic analysis using machine learning (ML) module, this stage involves preprocessing the obtained

dataset to extract relevant features while eliminating unnecessary ones. Feature selection techniques are applied to reduce dimensionality and improve the efficiency of the ML module. Various ML algorithms, such as Support Vector Machines (SVM), Naïve Bayes, Bayes Net, and Logistic Regression., are employed to train the ML module on the preprocessed dataset.

During the training process, the ML module learns to differentiate between legitimate and malicious network traffic based on the extracted features. Once trained, the ML model can analyze real-time live traffic and identify any suspicious patterns or deviations indicative of a potential DDoS attack. This enables prompt response and mitigation measures to be taken, effectively safeguarding the network infrastructure.

Stage of ML Module Updates:

Recognizing the dynamic nature of DDoS attacks, this stage focuses on the continuous improvement and adaptation of the ML module. Regular updates are essential to ensure the ML module effectively detects new and evolving DDoS assault techniques. These updates can be achieved through two primary approaches:

Retraining the ML model with fresh data: As new datasets become available or existing datasets are augmented, the ML module retrains to incorporate the latest knowledge. By exposing the model to recent attack patterns and network behaviors, we enhance its ability to accurately detect and classify DDoS attacks.

Algorithm enhancements: In addition to updating the training data, we explore opportunities to enhance the ML module's algorithms. By incorporating state-of-the-art techniques and leveraging advancements in ML research, we can improve the performance and efficiency of the module. Algorithmic adjustments may include optimizing hyperparameters, exploring ensemble methods, or adopting novel deep-learning architectures.

The stage of ML module updates ensures that our methodology remains robust and adaptive, capable of effectively countering the evolving landscape of DDoS attacks. By continually refining our detection capabilities, we aim to provide a reliable and comprehensive defense mechanism for network infrastructures.

By following this enhanced methodology, we can improve our approach's accuracy, efficiency, and adaptability in detecting and mitigating DDoS attacks, ultimately contributing to a more secure and resilient network environment.

V. PROPOSED MODEL

This section introduces the architecture of our proposed model ML approach for DDoS Detection. Our system architecture consists of three main stages: Traffic analysis using ML module stage to analyze the incoming

traffic. ML modules are used to classify the traffic into attack or non-attack traffic to detect DDoS based on the

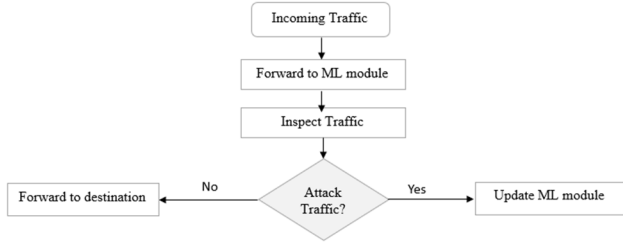


Fig. 3. proposed system architecture.

A. Dataset

To train our ML model, we used the CI-CDDoS2019 dataset [7]. It contains 12 different types of DDoS attacks, and each type is in a separate file. The attack types are UDP, SNMP, NetBIOS, LDAP, TFTP, NTP, SYN, Web DDoS, MSSQL, UDP-Lag, DNS, and SSDP DDoS-based attacks. The data is labeled as benign and DDoS. The CI-CDDoS2019 dataset is a publicly available dataset that was released in 2019 by the Canadian Institute for Cybersecurity. It contains network traffic data collected from a real-world network environment that was subjected to various DDoS attacks. The dataset also includes labeled data for each attack type, as well as benign traffic data, which can be used for training and testing ML-based DDoS detection models. The dataset provides a comprehensive set of attack scenarios and can be used to evaluate the performance of ML-based DDoS detection methods.

B. Data Preprocessing

The size of the dataset is 400,000 datapoints. To reduce the size before the Preprocessing phase, we removed the Flow ID, Source IP, Source Port, Destination IP, Destination Port, and Timestamp columns as they are unnecessary and have no effect on the result because these features are changed per case.

Furthermore, we replace the labels with numerical values, where 'Benign' is represented as 0 and 'DDoS' with 1.

Finally, we split the dataset first during data preparation and training to validation split. We did use a cross-validation set that is 15% the size of the training set. Since it helps to prepare the data for analysis and ensures that the ML model can accurately learn from it, data preprocessing is an important step in machine learning-based DDoS detection. The files' size is decreased to 400,000 data points in the first data preparation stage. This phase is essential to ensuring that the ML model can handle the data effectively and lower the computational complexity of the model. Several columns are eliminated from the dataset in the second stage since they are unique to each network and do not contain any data that can be used to identify DDoS. Flow ID, Source IP, Source Port, and Timestamp are among

the fields in this table. The dataset's dimensionality is decreased by removing these columns.

C. CNN-LSTM Model for DDoS Detection

Convolutional neural networks (CNNs) and long short-term memory (LSTM) are used in DDoS Detection. CNN is a deep learning model for processing data with a grid pattern, such as pictures. The first two layers are convolutional and max pooling to extract features, whereas the fully connected layer transfers the extracted features into the final output, which is classification. Convolutional, max-pooling, and fully connected layers are essential to CNN. Our CNN-LSTM architecture consists of 11 layers. Figure 4 shows a CNN-LSTM architecture.

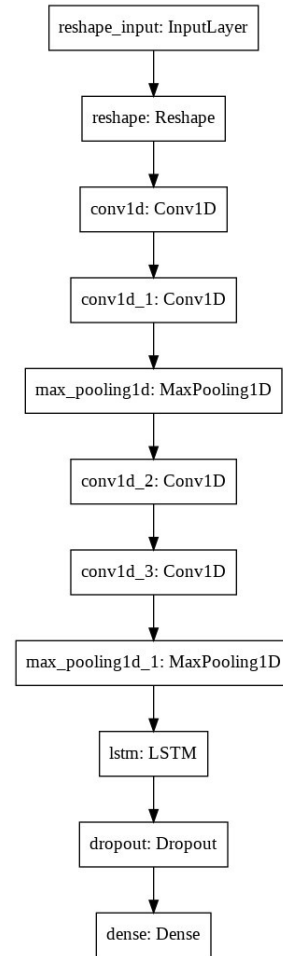


Fig. 4. CNN-LSTM architecture

The first layer only reshapes the incoming array to the required shape. Our input data has 67 features. So, the reshape layer makes an input shape of (67, 1), where 1 is the batch size. Next are the convolutional layers. They have filtered the filters. We have 64 filtered conv1D layer that moves along 67 data dimensions with a stride length of 2, so they each would have to move $(2*2*64) = 256$ trainable parameters. And each of them would again have more filters in the next layers. So, it increases exponentially. The max pooling layer is added to decrease the number of parameters to take only the max of the values in the window.

VI. RESULT AND DISCUSSION

Our study employed machine learning algorithms, including WEKA, to detect attacks. The CNN-LSTM model exhibited the highest accuracy among the various algorithms tested, achieving an impressive 99.51%. A comparison was made with other popular machine learning algorithms such as Naïve Bayes, SVM, Bayes Net, Random Forest, and Logistic Regression. To evaluate the accuracy of each algorithm, we analyzed two key factors: Correctly Classified Instances and Incorrectly Classified Instances across multiple types of attacks, including DNS, MSSQL, NETBOS, NTP, SSDP, SYN, TFTP, and UDPlag. By considering these factors, we could assess each algorithm's performance and identify the CNN-LSTM model as the superior approach, demonstrating its effectiveness in accurately detecting different types of attacks.

Figure 5 shows the accuracy of all machine learning algorithms except CNN-LSTM based on Correctly Classified Instances.

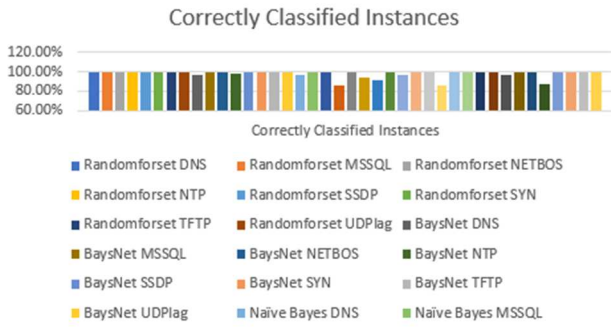


Fig. 5. Comparison of the accuracy of different ML algorithms.

Figure 6 shows the accuracy of all machine learning algorithms except CNN-LSTM based on Incorrectly Classified Instances.

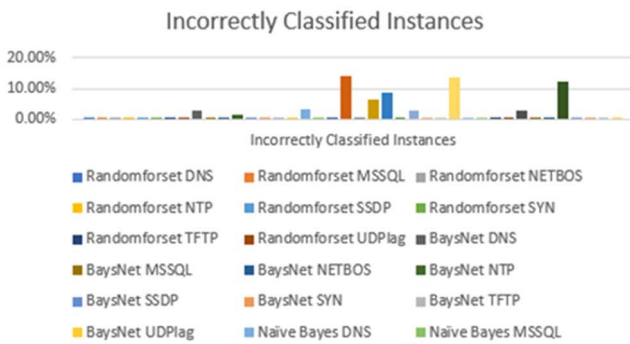


Fig. 6. Comparison of the inaccuracy of different ML algorithms.

To summarize the charts, the average of each algorithm over the different attacks was taken and compared with the CNN-LSTM results in Figure 7.

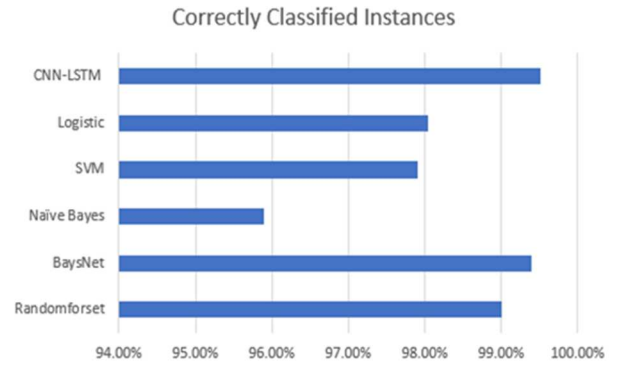


Fig. 7. Comparison of the accuracy of different ML algorithms with CNN-LSTM.

In our analysis, depicted in Figures 5, 6, and 7, we comprehensively compared all five algorithms across the eight previously mentioned attacks. To facilitate comparison, we calculated the average accuracy for each algorithm and juxtaposed them with the results obtained by the CNN-LSTM model. Notably, our findings revealed that the CNN-LSTM model consistently exhibited higher accuracy than all other algorithms. These results further underscore the superior performance and effectiveness of the CNN-LSTM model in accurately detecting and classifying the various types of attacks under consideration.

Table 2 shows the difference in the accuracy of detecting attacks between machine learning and our proposed model CNN-LSTM model.

TABLE I. A COMPARISON BETWEEN THE MACHINE LEARNING ALGORITHM AND THE PROPOSED MODEL

ALGORITHMS	ACCURACY
NAIVE BAYES	96%
SVM	97.40%
BAYES NET	97%
LOGISTIC REGRESSION	97%
RANDOM FOREST	99.01
CNN-LSTM	99.51%

VII. CONCLUSIONS

In conclusion, our proposed CNN-LSTM model has achieved an impressive accuracy of 99.51%, outperforming various machine learning algorithms, including Naïve Bayes, SVM, Bayes Net, and Logistic Regression. This outcome underscores the potential of our CNN-LSTM model as a highly effective approach for DDoS attack detection. The success of the CNN-LSTM model can be attributed to its unique ability to leverage the strengths of Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks. CNNs excel at processing spatial information, such as network traffic data, while LSTMs are adept at capturing temporal dependencies within

data, making them an ideal choice for analyzing time series data. By integrating these two architectures, our CNN-LSTM model effectively analyzes both the spatial and temporal aspects of network traffic data, resulting in significantly improved accuracy in detecting DDoS attacks.

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